Xianghui Shi

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Research Interests	Dynamical systems and ergodic theory, with a focus on thermodynamic formalism, statistical properties, and spectral rigidity of billiards; quasi-conformal geometry

Appointments

University of California, Los Angeles 2026–present
Hedrick Assistant Adjunct Professor, Department of Mathematics

Education

Peking University 2021–2026
Ph.D. in Mathematics, Advisor: Zhiqiang Li
Thesis: *Thermodynamic Formalism of Expanding Thurston Maps: Subsystems, Statistical Properties, and Computability*

Peking University 2017–2021
B.S. in Mathematics

Visiting Positions

Institute of Science and Technology Austria (ISTA) June–August 2025
Predoctoral Visiting Scientist, Host: Vadim Kaloshin

Publications

1. Thermodynamic formalism for subsystems of expanding Thurston maps and large deviations asymptotics (with Zhiqiang Li and Yiwei Zhang), **Proc. Lond. Math. Soc.** (3) 130, (2025), e70019. 77 pages. arXiv:2312.15822. PDF.
2. Thermodynamic formalism for subsystems of expanding Thurston maps II (with Zhiqiang Li), **J. Lond. Math. Soc.** (2), to appear. 46 pages. arXiv:2404.07247. PDF.

Preprints

3. Infinitesimal dynamical spectral rigidity of the Bunimovich stadium. arXiv:2609.16241. 42 pages.
4. Computable thermodynamic formalism (with Ilia Binder, Qiandu He, Zhiqiang Li). Submitted. arXiv:2512.01259. 41 pages. PDF.
5. Counting prime orbits in shrinking intervals for expanding Thurston maps (with Zhiqiang Li). Submitted. arXiv:2511.22601. 33 pages. PDF.
6. Entropy density and large deviation principles without upper semi-continuity of entropy (with Zhiqiang Li). Submitted. arXiv:2406.01712. 59 pages. PDF.

Talks and Presentations

Invited Seminar Talks

- *Ergodic theory of expanding Thurston maps and their subsystems*, The Budapest–Wien Dynamics Seminar, University of Vienna, Vienna, Austria, July 4, 2025.
- *Ergodic theory of expanding Thurston maps and their subsystems*, Quasiworld Seminar, online, May 7, 2025.
- *Ergodic theory of expanding Thurston maps and their subsystems*, Wuhan University, Wuhan, China, March 18, 2025.
- *Ergodic theory of expanding Thurston maps and their subsystems*, Seminar in Dynamical Systems, Institute of Science and Technology Austria, Klosterneuburg, Austria, December 3, 2024.

Invited Conference Talks

- *Ergodic theory of expanding Thurston maps and their subsystems*, Special Session on Transfer Operator and Thermodynamic Formalism, International Workshop on Operator Theory and its Applications (IWOTA 2026), Université Laval, Québec City, Canada, August 4, 2026.
- *Entropy density and large deviations without upper semi-continuity of entropy*, Symposium on Young Talent Support Project of the Chinese Mathematical Society, Shandong University, Jinan, China, April 25, 2025.

Contributed Talks

- *Statistics of prime orbits for expanding Thurston maps*, ISTA Summer School in Dynamical Systems, Strobl, Austria, September 26, 2025.
- *Ergodic theory of expanding Thurston maps and their subsystems*, Scuola Matematica Interuniversitaria Course on Dynamics of Billiards, Palazzone, Cortona, Italy, June 30, 2025.
- *Entropy density and large deviation principles without upper semi-continuity of entropy*, The Second National Doctoral Forum on Frontiers of Mathematics, Peking University, Beijing, China, October 31, 2024.

Poster and Short Presentations

- *Entropy density and large deviation principles without upper semi-continuity of entropy*, Conference on Holomorphic Dynamics and Fractal Geometry, Beijing International Center for Mathematical Research, Peking University, Beijing, China, January 14, 2025.
- *Ergodic theory of subsystems of expanding Thurston maps*, Conference on Dynamical Systems and Fractal Geometry (Lightning Talk), University of North Texas, Denton, USA, May 15, 2024.

Other Conferences Attended

- Fields-CNAM Nonlinear Days 2026, Fields Institute for Research in Mathematical Sciences, Toronto, Canada, September 14–18, 2026.
- Workshop "Spectral problems for planar domains and billiards", Erdős Center, Budapest, Hungary, November 28–30, 2024.

- Quasiworld Workshop, University of Helsinki, Helsinki, Finland, August 14–18, 2023.
- Advanced Studies Institute in Analysis on Fractal Spaces and Dynamical Systems, Urgench State University, Urgench, Uzbekistan, August 4–9, 2023.
- Conference "Around the Mandelbrot set: A conference celebrating the 60th birthday of Mitsuhiro Shishikura", Kyoto University, Kyoto, Japan, May 29–June 2, 2023.

Teaching Experience

University of California, Los Angeles, Department of Mathematics 2026–present
Instructor

- Math 32B: Calculus of Several Variables Fall 2026

Peking University, School of Mathematics 2021–2024
Graduate Teaching Assistant

- Mathematical Analysis III Fall 2024, Fall 2023, Fall 2022
- Mathematical Analysis II Spring 2022, Spring 2023
- Mathematical Analysis I Fall 2021

Awards and Honors

- Principal Scholarship, Peking University, 2025–2026
- BYD Graduate Scholarship, 2025
- College Award, Peking University, 2024–2025
- Young Elite Scientists Sponsorship Program for PhD Students, China Association for Science and Technology, 2024
- College Award, Peking University, 2023–2024
- BICMR Huaixin Scholarship (Outstanding Undergraduate Researchers), Peking University, 2021
- Academic Excellence Award, Peking University, 2018–2019
- 3rd Prize Scholarship, Peking University, 2018–2019
- Merit Student, Peking University, 2017–2018